Automated UI Testing for an E-Commerce Website

2. Requirements Gathering

To successfully develop and execute the automated UI test suite, the following requirements have been identified:

2.1. Environment and Technical Requirements

- **Programming Language:** Installation of a compatible programming environment (e.g., Python or Java).
- **Testing Framework:** Integration of the Selenium WebDriver library.
- **Browser Drivers:** Appropriate web browser drivers (e.g., ChromeDriver for Google Chrome) configured in the system PATH.
- **IDE:** A suitable Integrated Development Environment (e.g., VS Code, IntelliJ IDEA, or PyCharm) for script development and execution.

2.2. Functional Testing Requirements (Target User Journeys)

The automated test suite must cover the following core user workflows on the selected target e-commerce website (e.g., Sauce Labs "Swag Labs" demo site):

- **Authentication Flow:** Verification of the login mechanism using valid credentials and ensuring successful redirection to the main product page.
- **Product Search and Selection:** Ability to locate specific items and successfully add them to the shopping cart.
- **Cart Management:** Verification that the shopping cart dynamically updates (e.g., item counter assertions) when items are added or removed.
- **Checkout Process:** Navigation through the checkout pipeline, including the entry of billing/shipping details and final order confirmation.

2.3. Test Data Requirements

- **User Credentials:** Valid (and potentially invalid for negative testing) usernames and passwords.
- **Dummy Customer Information:** Synthetic data for first name, last name, and postal codes to complete the checkout verification without exposing real personal data.